

AM CVn: Testing a Superhump Interpretation

TODO: Team Members

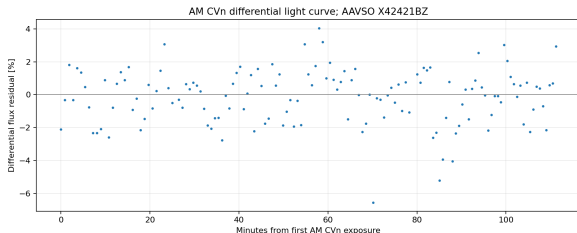
PHYSICS 100 Final Project

Scientific Goal

- AM CVn is an ultracompact helium-accreting binary.
- Literature periods: orbital 1028.7322 s, positive superhump 1051.2 s, strong harmonic near 525.6 s.
- Question: can our 2026 May 20 photometry tighten those constraints?
- Rule: separate free-period constraints from literature-conditioned fits.

Observations

- 140 V-band frames.
- 35, 40, and 45 s exposures.
- 138 good points after rejecting two artifact frames.
- Baseline: 111.20 min.
- Covers 6.35 cycles of a 1051.2 s superhump.



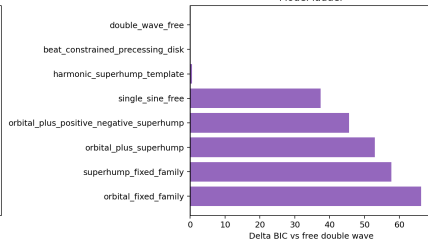
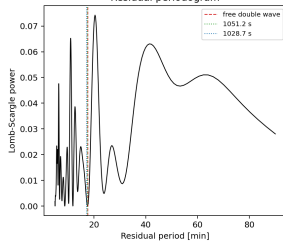
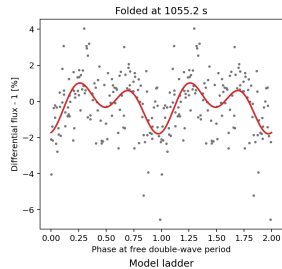
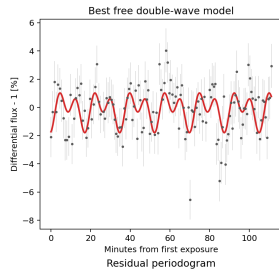
Reduction

- Bias, dark, and star-masked flat-field calibration.
- AAVSO sequence X42421BZ used for target and comparison-star geometry.
- Fluxes converted to rates before combining different exposure times.
- Comparison stars normalized by their own median before forming the ensemble.
- Robust target scatter: 1.51 percent; median formal error: 0.90 percent.

Model Ladder

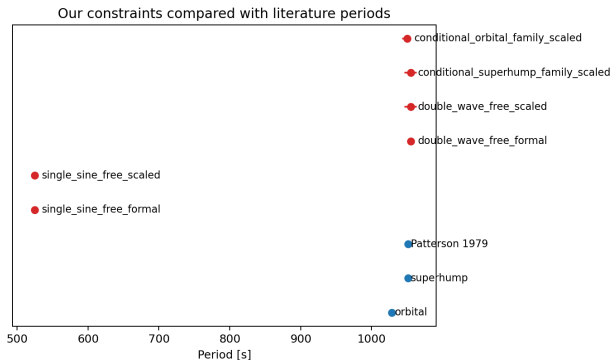
Model	Period	ΔBIC	CV RMS
Free double wave	1055.21 s	0.00	1.52%
Beat-constrained disk	1055.22 s	0.00	1.52%
1051.2 s + harmonic	fixed	0.55	1.53%
Free single sine	525.06 s	37.44	1.57%
Orbital + superhumps	fixed	45.55	1.65%
Superhump only	1051.2 s	57.67	1.62%
Orbital only	1028.73 s	66.18	1.65%

Best Physical Interpretation



Constraints

- Free single sine: 525.06 ± 4.27 s.
- Free double wave: 1055.21 ± 8.58 s.
- Conditional superhump family:
 1055.22 ± 8.64 s.
- These are consistent with the 1051.2 s superhump family.



Validation

- Multi-harmonic Lomb-Scargle recovers the double-wave fundamental.
- Beat-constrained model implies $P_{\text{prec}} \approx 11.39$ hr, inferred rather than directly observed.
- Leave-one-comparison-star-out period span: 5.2 s.
- Outlier clipping period shift: 0.02 s.
- Residual periodogram has no significant remaining peak.
- Systematics terms lower RMS to 1.41%, but worsen BIC/CV.
- Robust losses shift the answer to 1057–1059 s and worsen RMS.
- First/second halves differ by 22.8 s.

Could It Be Something Else?

- Broadband photometry alone cannot prove the object class.
- Literature spectroscopy shows helium accretion signatures and an S-wave orbital feature.
- A non-binary pulsator would need a spectrum without accretion features and a different multi-period pattern.
- The decisive test is time-resolved spectroscopy, not another photometric periodogram.

Conclusion

- We detect AM CVn short-timescale variability.
- The 525 s single-sine signal is best interpreted as a harmonic.
- The double-wave and 1051.2 s superhump-template fits support the known superhump/harmonic structure.
- Detrending, robust likelihoods, and comparison subsets do not improve the constraint.
- We do not tighten the literature periods with this dataset.
- A longer multi-night baseline is the path to a stronger constraint.